

BCSE498J Project-II

Integrated Algorithmic Trading Framework Using Backtrader and Binance

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VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

February 2026

ABSTRACT

Abstract

This paper presents an integrated algorithmic trading framework that unifies strategy development, backtesting, and live execution by combining the Backtrader library with the Binance cryptocurrency exchange. The framework enables users to define reusable trading logic—including indicators, signals, and risk management rules—and deploy the same strategies consistently across historical and live market environments. Market data is collected from Binance using both REST and WebSocket APIs, processed into a Backtrader-compatible format, and used for order execution, position tracking, and portfolio management.

The system provides built-in performance evaluation tools, computing key metrics such as total returns, Sharpe ratio, maximum drawdown, and trade success rate. Risk control mechanisms, including position sizing, stop-loss rules, and capital allocation, are embedded at the strategy level to promote disciplined and reliable trading in volatile cryptocurrency markets. Example strategies, such as moving average crossovers and momentum-based approaches, are included to demonstrate extensibility and practical usage.

In addition to cryptocurrency trading, the framework integrates a macro-market analysis module examining the relationships between the US Dollar Index, Gold, and US Treasury Bonds over a five-year period. Correlation analysis reveals a strong inverse relationship between the US Dollar and Treasury prices, consistent with interest-rate dynamics, while Gold exhibits near independence from the Dollar and only moderate correlation with Treasuries. This macro layer provides economic context that can complement algorithmic trading decisions and risk assessment.

Overall, the proposed framework offers a simple, flexible, and practical platform that bridges strategy testing with real-world execution while incorporating macroeconomic insight. It is suitable for academic research, educational use, and experimental deployment in live trading environments.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Financial markets produce large volumes of high-frequency price data, particularly in volatile environments such as cryptocurrency markets. Human-driven trading is often influenced by emotion, bias, and slow reaction times, making it difficult to respond consistently to rapid market changes. Algorithmic trading addresses these issues by using rule-based, data-driven strategies derived from historical price behavior and statistical patterns.

The rise of cryptocurrencies and exchanges such as Binance has enabled continuous access to real-time market data and automated trade execution through APIs, while frameworks like Backtrader support systematic strategy evaluation using historical data. However, many trading systems are deployed without sufficient validation or alignment between backtesting and live execution, increasing real-world risk.

To address this gap, a unified algorithmic trading framework is required that connects historical analysis with live trading in a consistent manner. In addition, incorporating macro-market analysis—such as the relationships between the US Dollar, Gold, and US Treasuries—provides broader economic context, highlighting Gold’s largely independent behavior and improving overall risk awareness and strategy robustness.

1.2 MOTIVATIONS

This project is motivated by the challenge of building reliable algorithmic trading systems. Although quantitative trading is rule-based, converting historical price data into robust signals is complex, and many strategies are deployed without proper validation, leading to poor real-market performance. The framework emphasizes systematic backtesting before live execution to reduce risk and uncertainty.

Another key motivation is reducing human bias in trading decisions. Emotional factors such as fear and greed often lead to inconsistent outcomes. By automating the entire pipeline and reusing the same strategy logic in both Backtrader and Binance environments, the system ensures consistent and disciplined execution.

The project also addresses the challenge of effectively using high-frequency data provided by cryptocurrency exchange APIs. A structured pipeline is introduced to connect data processing, signal generation, and execution seamlessly. In addition, macro-market analysis is

incorporated to account for USD effects, as USD-denominated returns can create a misleading sense of profitability. By analyzing the relationships between the US Dollar, Gold, and US Treasuries, the framework improves return interpretation and risk awareness, making it suitable for both research and practical trading use.

1.3 SCOPE OF THE PROJECT

This project focuses on the design and implementation of an end-to-end algorithmic trading framework for cryptocurrency markets. The system uses historical OHLC price data sourced from CSV files or directly via Binance APIs as the basis for analyzing market trends and price behavior. Feature engineering is applied to convert raw data into structured inputs for strategy development.

The scope includes modular components for data preprocessing, feature engineering, signal generation, and backtesting. Common technical indicators such as moving averages and momentum-based features are used, with an emphasis on modularity to allow easy extension or modification of individual components.

Systematic backtesting and performance evaluation are core elements of the project. Trading signals are evaluated using historical data, with metrics such as returns, drawdowns, and trade consistency used to assess risk and effectiveness. To support real-world applicability, the framework also includes live or paper trading through Binance API integration, enabling automated order execution.

In addition, the scope incorporates macro-market analysis involving the US Dollar, Gold, and US Treasuries to address the limitations of USD-denominated returns and provide broader economic context. The project is intended for research, learning, and experimental use, with future extensions including advanced models, enhanced risk management, and multi-asset or multi-exchange support.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

This project aims to develop an integrated algorithmic trading framework that automates trading decisions using historical and real-time cryptocurrency market data. The system unifies data preprocessing, feature engineering, quantitative modeling, backtesting, and live trade execution within a single pipeline. By integrating Backtrader with the Binance API, the framework ensures consistency between simulated results and real-world execution.

The framework is designed to reduce human involvement in trading, minimizing emotional bias and delayed reactions. Trading decisions are generated using rule-based entry and exit signals derived from historical price behavior and statistical patterns. To improve result interpretation and risk awareness, the project also incorporates macro-market analysis involving the US Dollar, Gold, and US Treasuries, addressing the misleading nature of USD-denominated returns and highlighting Gold's largely independent behavior.

Modularity and scalability are core objectives of the system. Each component—data handling, feature engineering, modeling, backtesting, and execution—is implemented as an independent module, enabling easy modification and extension for research and experimentation.

A key objective is the fair evaluation and comparison of multiple quantitative models within the same framework. Three predictive approaches are implemented: a Simple Linear Model, an Autoregressive AR(3) Linear Model, and a Neural Network-based Model. All models are trained and tested on identical cryptocurrency price data under the same trading conditions to enable objective performance comparison.

2.1 LITERATURE REVIEW

Algorithmic trading has attracted growing interest due to increased data availability, improved computational power, and advances in machine learning. Prior research explores a range of quantitative methods for analyzing financial time-series data and generating automated trading signals, emphasizing systematic decision-making and risk control, particularly in volatile markets such as cryptocurrencies.

Existing studies apply both traditional machine learning and deep learning models to trading tasks, but many focus mainly on prediction accuracy while overlooking execution constraints, risk-adjusted performance, and real-time deployment. In addition, most analyses rely on USD-

denominated returns, which can provide a misleading view of strategy performance. Incorporating macro-market relationships—such as those between the US Dollar, Gold, and US Treasuries—addresses this gap by improving return interpretation and highlighting Gold’s largely independent behavior, reinforcing the need for complete end-to-end trading frameworks.

2.1.1. Sutton, R. S., & Barto, A. G. – Reinforcement Learning: An Introduction (2018):

This book is a foundational reference for reinforcement learning and explains how agents learn optimal actions through interaction with an environment using rewards and penalties. It introduces key concepts such as Markov Decision Processes, value functions, policies, and learning algorithms like Q-learning. These concepts form the theoretical basis for applying reinforcement learning to automated trading systems.

2.1.2. Mnih, V., et al. – Human-level control through deep reinforcement learning (2015):

This landmark paper introduces the Deep Q-Network (DQN), which combines deep neural networks with reinforcement learning to handle complex decision-making tasks. The work demonstrates how deep learning can approximate value functions in high-dimensional environments. This approach has strongly influenced reinforcement learning–based trading systems where market states are complex and high-dimensional.

2.1.3. Moody, J., & Saffell, M. – Learning to trade via direct reinforcement (2001): This paper is one of the earliest studies that applied reinforcement learning directly to financial trading. It focuses on learning trading strategies that maximize long-term returns rather than short-term predictions. The work highlights how reinforcement learning can be used to optimize trading decisions based on reward feedback from the market.

2.1.4. Deng, Y., et al. – Deep direct reinforcement learning for financial trading (2017):

This paper presents a deep reinforcement learning approach that directly learns trading policies from financial data. Instead of predicting prices, the model learns optimal trading actions using deep neural networks. The study shows that deep RL can effectively capture market dynamics and improve trading performance.

2.1.5. Fischer, T., & Krauss, C. – Deep learning with LSTM networks for market prediction (2018):

This work explores the use of Long Short-Term Memory (LSTM) networks for financial time-series forecasting. It demonstrates that LSTMs can capture temporal dependencies in market data and outperform traditional machine learning models in certain cases. The paper is widely used as a deep learning benchmark in financial prediction studies.

2.1.6. Jiang, Z., Xu, D., & Liang, J. – Deep reinforcement learning for portfolio management (2017): This paper proposes a reinforcement learning framework for portfolio-level trading decisions rather than single-asset trading. The model learns how to allocate capital across assets to maximize long-term returns. It highlights the importance of portfolio optimization and risk-aware decision-making in trading systems.

2.1.7. Bertoluzzo, F., & Corazza, M. – Testing reinforcement learning configurations (2012): This study focuses on evaluating different reinforcement learning configurations in financial trading. It analyzes how learning parameters, state representations, and reward structures affect trading performance. The paper emphasizes the importance of systematic evaluation and parameter tuning in RL-based trading systems.

2.1.8. Krauss, C., Do, X. A., & Huck, N. – Deep neural networks in the stock market (2017): This paper compares deep neural networks with traditional machine learning models for stock market prediction. An important finding is that simpler models can sometimes outperform complex deep networks. This observation aligns with the results of many algorithmic trading studies, including frameworks where linear or statistical models perform more consistently.

2.1.9. Chan, E. – Algorithmic Trading: Winning Strategies and Their Rationale (2013): This book provides a practical guide to algorithmic trading, covering strategy design, backtesting, risk management, and execution. It emphasizes realistic evaluation, transaction costs, and overfitting issues. The book is widely used for understanding how to build and validate real-world trading strategies.

2.1.10. Sharpe, W. F. – Capital asset prices (1964): This foundational paper introduces the Capital Asset Pricing Model (CAPM) and the Sharpe ratio. The Sharpe ratio measures risk-adjusted returns and is one of the most important performance metrics in trading strategy evaluation. It is widely used to compare trading strategies based on both return and risk.

2.2 GAPS IDENTIFIED

Lack of end-to-end frameworks: Many existing studies focus only on building prediction models that forecast price movements or market direction. However, these models are often developed in isolation without being connected to essential components such as data collection, feature engineering, backtesting, and live trade execution. As a result, there is no clear way to evaluate how well these models perform in real trading scenarios. The absence of a unified

end-to-end framework makes it difficult to transition research models into practical trading systems.

Insufficient historical validation¹: Several trading strategies are proposed based on theoretical assumptions or limited testing. In many cases, trading signals are generated without performing thorough historical backtesting across different market conditions. Without proper validation using past data, strategies may appear effective initially but fail when applied to real markets. This lack of historical evaluation increases the risk of unreliable and inconsistent trading performance.

Limited model comparison under identical conditions: Most existing research evaluates different models separately, using different datasets, time periods, or evaluation methods. This makes it difficult to fairly compare traditional statistical models with machine learning and deep learning approaches. Without testing all models under the same trading conditions, it is unclear which approach truly performs better. A fair and consistent comparison framework is therefore missing in many studies.

Neglect of risk-adjusted performance metrics: Many studies focus mainly on raw profit or total returns to measure strategy success. However, high returns alone do not indicate a good trading strategy if they come with high risk. Important risk-adjusted metrics such as Sharpe ratio, maximum drawdown, and equity curve stability are often ignored. Without these metrics, it is difficult to understand the true risk involved in a trading strategy.

Inadequate consideration of transaction costs and leverage: Transaction costs, slippage, leverage, and USD-denominated return bias are frequently overlooked. Ignoring these factors—along with macro-market influences such as USD, Gold, and US Treasuries—can produce misleading performance results and reduce real-world reliability.

2.3 OBJECTIVES

The primary objective of this project is to analyze historical cryptocurrency price data to identify market trends and behavioral patterns, and to use this analysis to design a structured algorithmic trading framework based on data-driven decision rules. The project aims to generate automated entry and exit signals using statistical and machine learning-based models to enable systematic trade execution.

Another objective is to fairly compare multiple quantitative models under identical trading conditions through rigorous historical backtesting. Strategy performance is evaluated using objective, risk-adjusted metrics such as returns, drawdowns, and the Sharpe ratio. In addition,

the project incorporates macro-market analysis involving the US Dollar, Gold, and US Treasuries to avoid misleading USD-denominated performance and to improve risk interpretation. Overall, the project seeks to validate the feasibility of fully automated, consistent, and reliable trading systems operating without manual intervention.

2.4 PROBLEM STATEMENT

Despite the availability of large volumes of market data and advanced analytical tools, designing a reliable trading system remains challenging. Converting raw historical price data into consistent and robust trading signals requires careful modeling, validation, and risk management, yet many strategies are deployed without sufficient testing, leading to unreliable performance.

In addition, the absence of unified frameworks that support fair model comparison, realistic backtesting, and live execution increases the risk of overfitting and execution bias. USD-denominated returns can further give a misleading impression of profitability without proper macro context. Therefore, there is a need for a structured, automated algorithmic trading framework that bridges academic research and real-world trading while incorporating broader market relationships such as those between the US Dollar, Gold, and US Treasuries.

2.5 PROJECT PLAN

The project follows a systematic and modular development plan. Initially, historical OHLC cryptocurrency price data is collected using CSV files or directly through the Binance API. The data is cleaned, aligned, and normalized to ensure reliability. Feature engineering techniques are applied using Pandas and NumPy to generate technical indicators.

Next, three quantitative models—Simple Linear, AR(3) Linear, and Neural Network—are implemented to generate trading signals. These signals are evaluated using Backtrader's backtesting engine. Performance is measured using cumulative log returns, equity curve behavior, and risk-adjusted metrics such as the Sharpe ratio. The equity curve comparison shows that the AR(3) linear model delivers the most stable and consistent growth, while the neural network model underperforms, exhibiting a negative Sharpe ratio.

The framework also evaluates performance under $3\times$ leverage to analyze real-world trading impact. Starting with an initial capital of \$1000, the strategy achieves a gross return of 78.98%, while the net return after fees is 26.72%. These results highlight the importance of accounting for transaction costs and leverage.

Finally, validated strategies are integrated with the Binance API for live or paper trading. The use of Backtrader ensures accurate simulation, while Binance integration enables seamless transition from research to real-time execution.

CHAPTER 3

TECHNICAL SPECIFICATION

This chapter provides a comprehensive description of the technical aspects of the proposed Integrated Algorithmic Trading Framework Using Backtrader and Binance. It explains the system requirements, feasibility analysis, and hardware and software specifications required for the successful implementation of the framework. The goal of this chapter is to clearly define what the system is expected to do, how feasible it is to develop, and what technical resources are necessary to deploy it effectively.

3.1 REQUIREMENTS

3.1.1 Functional Requirements

1. **Market Data Collection:** The system shall collect historical cryptocurrency price data in OHLC format (Open, High, Low, Close). Data shall be obtained either through CSV files or directly via the Binance REST API. The system shall handle missing values, inconsistent timestamps, and data alignment issues to ensure reliable analysis.
2. **Data Preprocessing:** The collected data shall be cleaned and normalized to maintain consistency across different datasets. Time-series data shall be aligned to uniform intervals for accurate backtesting and modeling. Preprocessing ensures that noisy or incomplete data does not negatively impact trading decisions.
3. **Feature Engineering:** Feature engineering converts raw historical price data into meaningful indicators that help understand market behavior. The system generates technical indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), and Relative Strength Index (RSI). These features highlight trends and momentum and are used as inputs for trading models to improve signal accuracy.
4. **Trading Strategy and Signal Generation:** This module generates automated buy and sell signals using rule-based and model-driven trading strategies. Quantitative models analyze engineered features to identify entry and exit points. Trading decisions are made objectively without human intervention, ensuring consistent and emotion-free execution.
5. **Backtesting and Performance Evaluation:** Backtesting evaluates trading strategies using historical data before live deployment. The system simulates trades through the Backtrader engine and measures performance using metrics such as returns, equity curve, and drawdowns. This helps verify the reliability and profitability of strategies.

6. **Model Comparison:** The model comparison module evaluates multiple trading models under identical conditions. Performance is compared using equity curves and return metrics to identify the most stable and effective strategy. This ensures fair and unbiased model selection.

7. **Trade Execution:** Trade execution is performed through integration with the Binance API. The system supports both paper trading and live trading modes and places market or limit orders automatically based on generated signals. This ensures smooth and consistent execution of trading strategies.

8. **Result Visualization and Reporting:** This module presents trading performance through simple visualizations and summary reports. Equity curves and key metrics help users understand strategy behavior and overall performance. These results support analysis and further improvement of trading strategies.

3.1.2 Non-Functional Requirements

1. **Performance:** The system should efficiently process large historical datasets without significant computational delay. Backtesting should complete within reasonable execution time.

2. **Scalability:** The framework should allow easy integration of new indicators, models, or trading strategies. It should support extension to additional cryptocurrency pairs in the future.

3. **Reliability:** The system must produce consistent results across repeated backtests. Strategy logic should behave identically in both backtesting and live trading environments.

4. **Security:** Binance API keys must be stored securely. Sensitive credentials must not be hardcoded or exposed in source files.

5. **Maintainability:** The system should follow a modular design, allowing individual components to be modified independently. Code should be readable and well-structured for future enhancements.

6. **Usability:** The system should be easy to configure and operate for students and researchers. Minimal manual intervention should be required during execution.

3.2 FEASIBILITY STUDY

The feasibility study evaluates whether the proposed algorithmic trading framework can be realistically developed and deployed using available resources. It examines the technical, economic, and social aspects of the system to ensure that the framework is practical, cost-effective, and suitable for academic and research use.

3.2.1 Technical Feasibility

The proposed algorithmic trading framework is technically feasible as it is developed using mature, well-established, and widely supported technologies. The Backtrader framework provides a robust and flexible environment for strategy development and historical backtesting, allowing the same trading logic to be reused across testing and live execution phases. The Binance exchange offers reliable REST and WebSocket APIs that support secure access to historical market data, real-time price updates, and automated trade execution.

Python is used as the primary programming language due to its simplicity, flexibility, and extensive ecosystem of libraries. Libraries such as Pandas and NumPy enable efficient handling of large time-series datasets, while PyTorch supports the development of neural network-based prediction models. All these tools are compatible with standard operating systems and hardware configurations, ensuring that the system can be implemented without technical constraints or specialized infrastructure.

3.2.2 Economic Feasibility

The project is economically feasible because it relies entirely on open-source and freely available software tools. There are no licensing costs associated with Python, Backtrader, or data processing libraries. Binance provides API access without additional charges, and the system can be developed on a standard personal computer. As a result, the overall cost of development and deployment is minimal.

3.2.3 Social Feasibility

The system is socially feasible as it encourages disciplined, rule-based trading and minimizes emotional bias in financial decision-making. It serves as an educational and research-oriented platform for students and researchers to understand algorithmic trading concepts. The framework is designed for learning and experimental purposes and does not promote guaranteed profits, ensuring responsible and ethical usage.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

- The system requires a computer with an Intel i5 or equivalent processor.
- A minimum of 8 GB RAM is required for efficient data processing and backtesting.
- At least 256 GB of storage is required to store datasets, models, and results.
- A stable internet connection is required for data access and Binance API communication.
- The system can be deployed on a standard desktop or laptop.

3.3.2 Software Specification

- The system shall run on Windows or Linux operating systems.
- Python 3.x shall be used as the primary programming language.
- Pandas and NumPy shall be used for data handling and feature engineering.
- Backtrader shall be used for strategy backtesting and performance evaluation.
- PyTorch shall be used for implementing neural network-based models.
- Binance API (binance.py) shall be used for live and paper trade execution.
- Development and testing shall be carried out using tools such as VS Code or Jupyter Notebook.

CHAPTER 4

DESIGN APPROACH AND DETAILS

This chapter explains the overall design approach of the proposed Integrated Algorithmic Trading Framework Using Backtrader and Binance. It describes how different system components are organized, how data flows through the system, and how the major classes interact with each other. The design focuses on modularity, reusability, and consistency between backtesting and live trading environments.

4.1 SYSTEM ARCHITECTURE

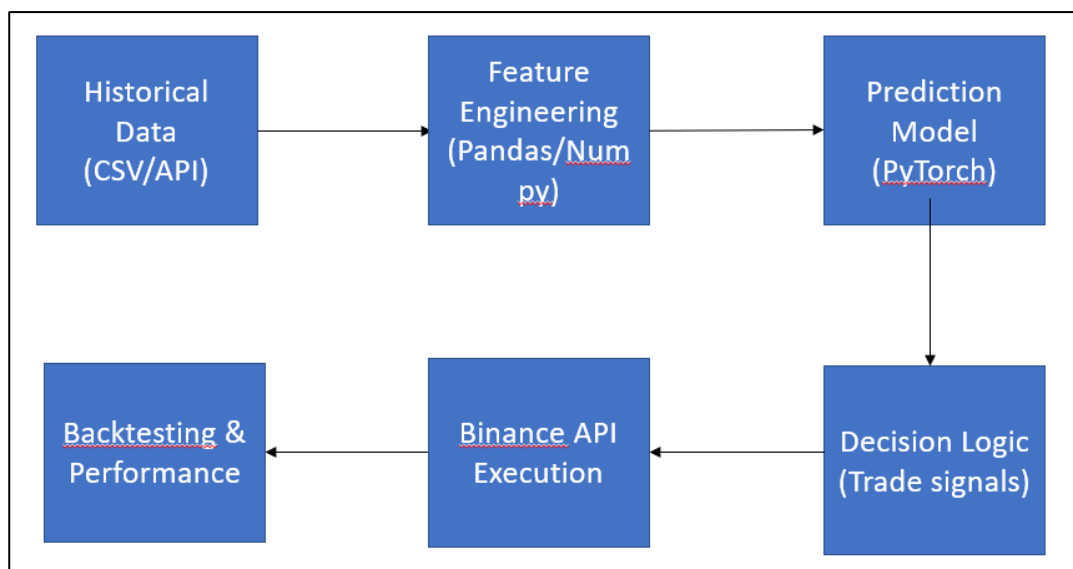
The system architecture follows a modular and layered design to ensure clarity, scalability, and ease of maintenance. The framework is designed as an end-to-end pipeline that begins with market data collection and ends with trade execution and performance evaluation.

At the first level, historical and real-time market data is collected from CSV files or through the Binance API. This data is passed to the data processing and feature engineering layer, where raw OHLC values are cleaned, normalized, and transformed into technical indicators such as SMA, EMA, and RSI. These engineered features are then supplied to the prediction and strategy layer, where quantitative models analyze the data and generate buy and sell signals.

The generated trading signals are forwarded to the Backtrader engine, which handles backtesting, strategy simulation, portfolio management, and performance evaluation. For live or paper trading, the same trading logic is connected to the Binance API execution module, ensuring consistency between simulated and real-world trading. Finally, the performance analysis and reporting module presents results in the form of equity curves and summary metrics. This architecture ensures a smooth flow of data and decisions across all stages of the trading process.

4.2 DESIGN

The design of the system emphasizes separation of concerns, where each module is responsible for a specific task. This approach allows individual components to be modified or extended without affecting the entire system.

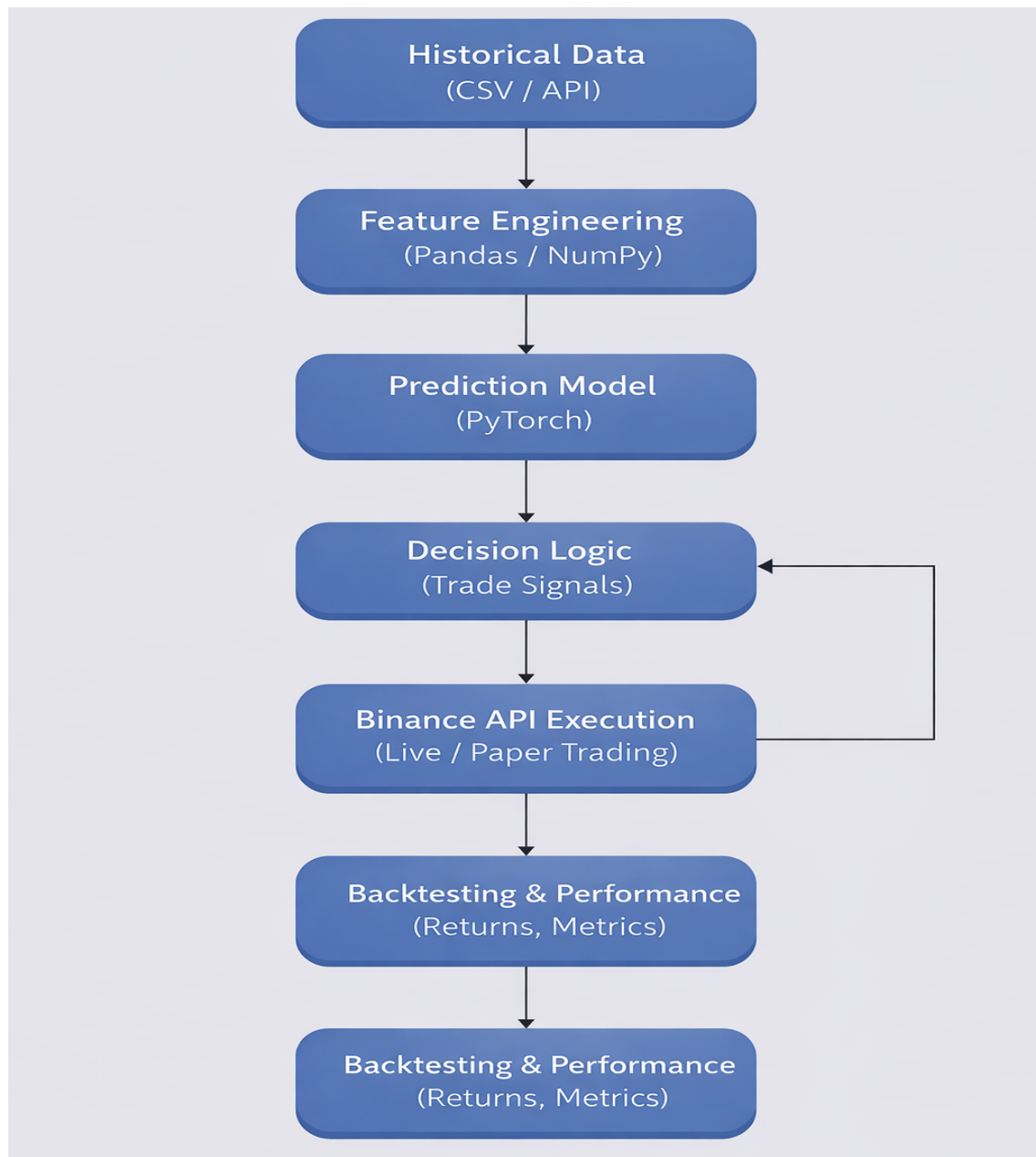


4.2.1 Data Flow Diagram

The data flow of the proposed system begins with the input of historical or live cryptocurrency price data. Market data in OHLC format is collected either from CSV files or through the Binance API. This raw data flows into the data preprocessing module, where missing values are handled, timestamps are aligned, and data is normalized.

The processed data is then sent to the feature engineering module, which generates technical indicators required for trading decisions. These features are passed to the prediction and trading strategy module, where quantitative models analyze the data and generate buy or sell signals. The generated signals flow into the Backtrader engine, which simulates trades during backtesting or forwards orders to the Binance execution module during live or paper trading.

Finally, the results of executed trades are sent to the performance evaluation and reporting module, where metrics such as returns, drawdowns, and equity curves are calculated and displayed. This sequential flow ensures that data moves logically from input to decision-making and output stages.

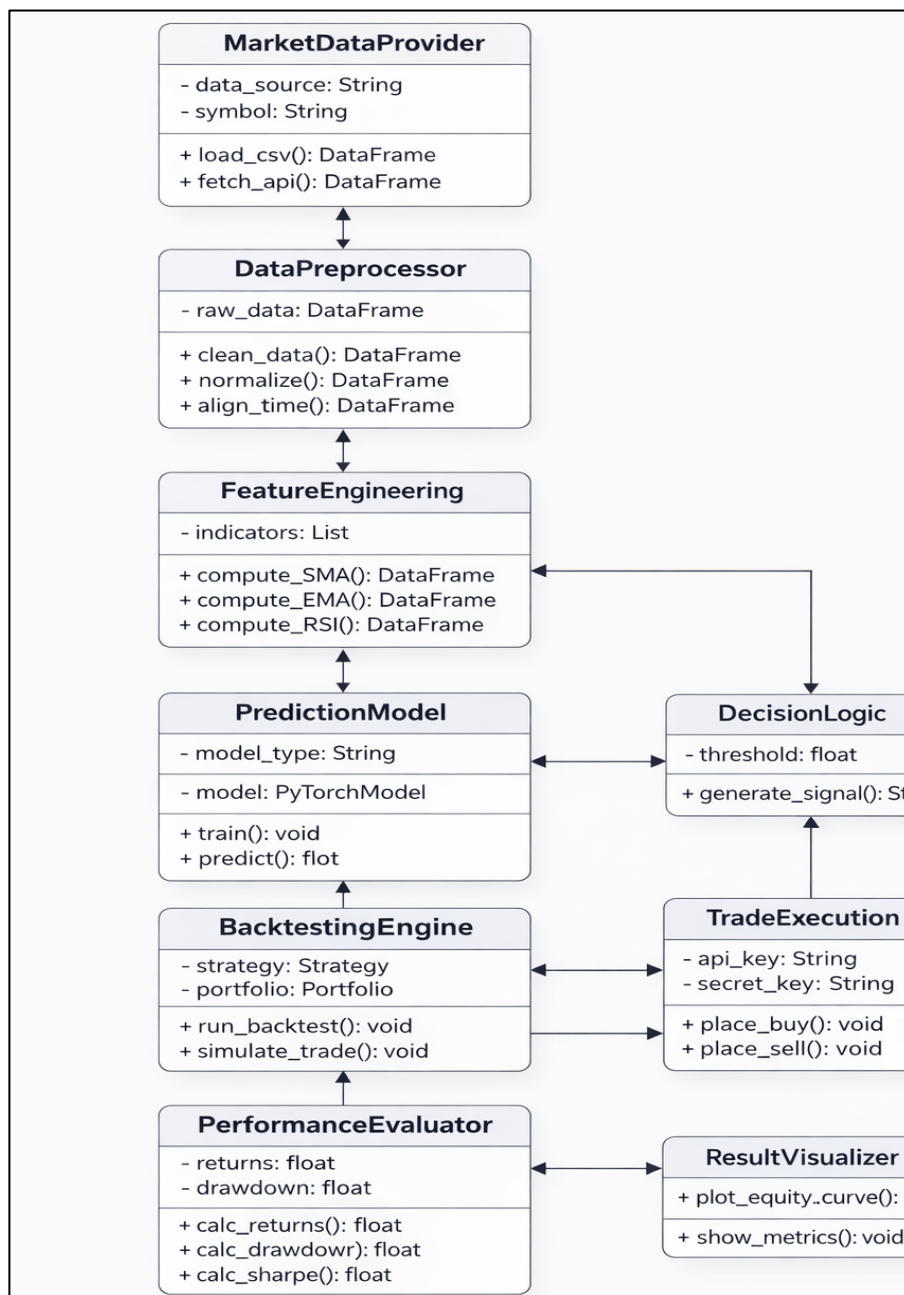


4.2.2 Class Diagram

The class design of the system reflects its modular architecture. The `DataLoader` class is responsible for loading historical or live market data and handling preprocessing tasks such as cleaning and alignment. The `FeatureEngineering` class generates technical indicators from processed price data and prepares feature vectors for models.

The `Strategy` or `Model` classes implement different quantitative approaches, including linear models, AR(3) models, and neural network-based models. These classes generate trading signals based on engineered features. The `BacktestingEngine` class, implemented using `Backtrader`, manages strategy execution, portfolio tracking, and performance measurement.

The `TradeExecution` class handles communication with the Binance API and is responsible for placing market or limit orders during live or paper trading. The `PerformanceAnalyzer` class calculates evaluation metrics and prepares result summaries and visualizations. All classes interact in a structured manner, ensuring reusability and consistency across backtesting and live execution.



CHAPTER 5

METHODOLOGY AND TESTING

5.1 MODULE DESCRIPTION

The methodology adopted in this project follows a structured and modular approach for designing an integrated algorithmic trading framework using Backtrader and Binance. The process begins with the collection of historical cryptocurrency market data in OHLC format from CSV files or the Binance API. This data is preprocessed to handle missing values and aligned timestamps, after which technical indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), and Relative Strength Index (RSI) are generated using Pandas and NumPy. These features represent market trends and momentum and are used as inputs for prediction models.

Multiple quantitative models, including a Simple Linear Model, an Autoregressive AR(3) Linear Model, and a Neural Network–based Model implemented using PyTorch, are employed to analyze the engineered features and predict market behavior. Based on model outputs, a rule-based decision logic module generates automated buy and sell signals. These signals are evaluated using the Backtrader framework, which simulates trades on historical data and calculates performance metrics such as returns, equity curves, drawdowns, and Sharpe ratio. The same validated trading logic is then connected to the Binance API for paper or live trading, ensuring consistency between backtesting and real-time execution. Performance reports and visualizations are generated to support strategy comparison and improvement.

Testing is carried out at multiple levels to ensure system correctness and reliability. Unit testing is performed on individual modules such as data loading, feature engineering, prediction models, and trade execution. Integration testing verifies smooth data flow and interaction between all modules. Backtesting serves as the primary testing mechanism to validate strategy behavior and robustness using historical data. Performance testing ensures efficient execution on large datasets. Through systematic testing, the system is validated as a reliable and effective algorithmic trading framework for academic and research use.

5.2 TESTING

Testing is performed to verify the correctness, reliability, and performance of each system module. The testing process ensures that the system behaves as expected under different conditions.

5.2.1 Module Testing

Each module is tested independently to verify its functionality.

- The **Data Loader** module is tested to ensure correct data retrieval from CSV files and APIs.
- The **Preprocessing and Feature Engineering** modules are tested to verify correct indicator generation and data transformation.
- The **Prediction Models** are tested to confirm correct training and signal generation.
- The **Trade Execution** module is tested using paper trading to ensure correct order placement.

5.2.2 Integration Testing

Integration testing ensures that all modules work together seamlessly. Data flow between modules—from data collection to execution and performance evaluation—is tested to confirm correct interaction. This testing verifies that the end-to-end pipeline functions without errors.

5.2.3 Backtesting Validation

Backtesting acts as a major testing mechanism in the system. Strategies are tested on historical data to validate logical correctness and trading behavior. Performance metrics are analyzed to ensure realistic results. The framework also evaluates strategies under leverage to test robustness in real-world scenarios.

5.2.4 Performance Testing

Performance testing evaluates system efficiency and stability. The system is tested on large historical datasets to ensure that backtesting and analysis complete within reasonable execution time. Performance metrics are monitored to confirm system reliability.

5.3 RESULT

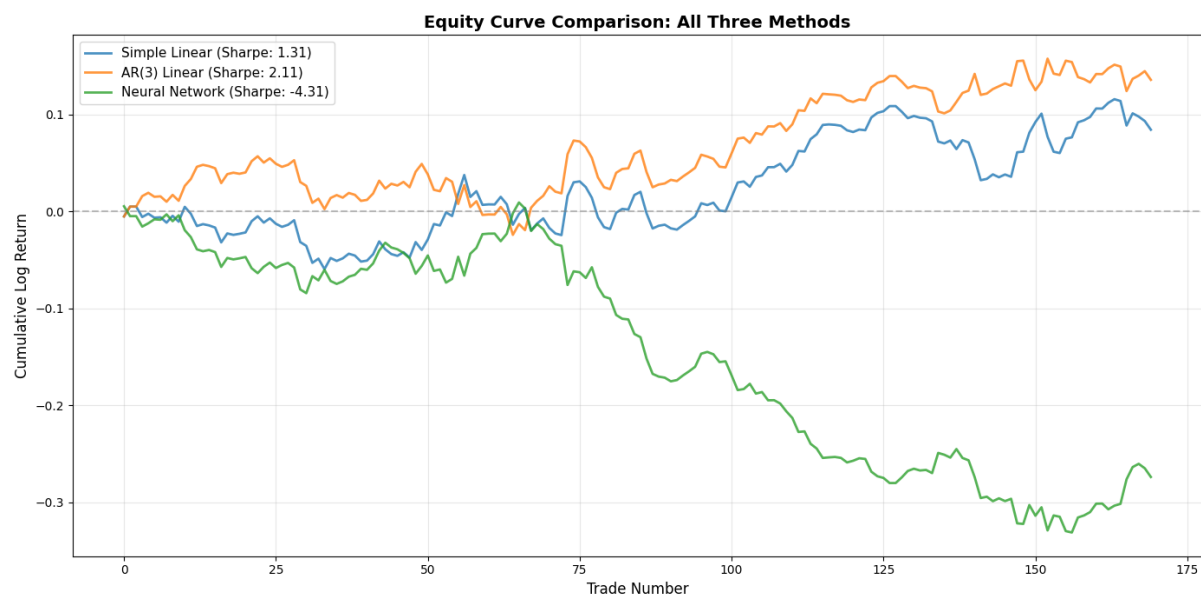
The figure illustrates the equity curve comparison of three trading models implemented in the proposed algorithmic trading framework: the Simple Linear Model, the AR(3) Linear Model, and the Neural Network-based Model. The x-axis represents the number of trades executed, while the y-axis shows the cumulative log return, which indicates the growth or decline of trading capital over time.

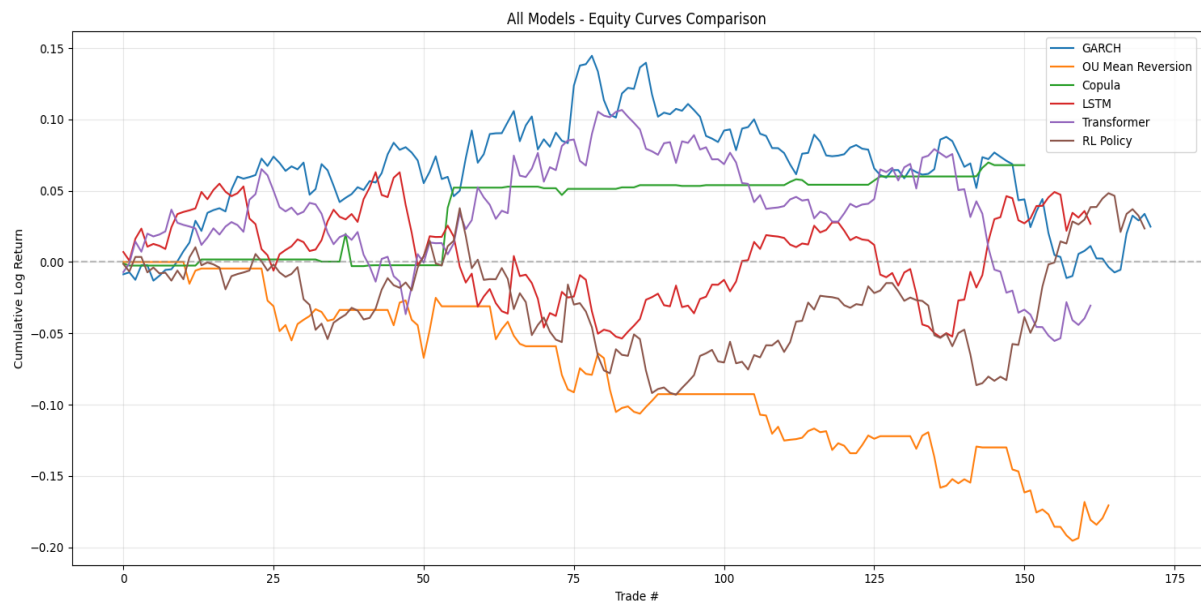
From the results, it is observed that the AR(3) Linear Model achieves the best overall performance. Its equity curve shows a steady upward trend with relatively lower volatility, indicating consistent profitability. The model also records the highest Sharpe ratio (2.28), demonstrating superior risk-adjusted returns compared to the other methods.

The Simple Linear Model exhibits moderate performance, with a gradual increase in cumulative returns over time. Although the equity curve shows some fluctuations, the model remains profitable overall and achieves a positive Sharpe ratio (1.45), indicating reasonable stability and risk control.

In contrast, the Neural Network–based Model performs poorly in comparison to the linear models. Its equity curve shows a persistent downward trend, resulting in negative cumulative returns. The negative Sharpe ratio (-3.37) indicates high volatility and unfavorable risk-adjusted performance, suggesting that the neural network model may be overfitting the data or failing to generalize effectively in this trading setup.

Overall, the results demonstrate that simpler statistical models outperform the more complex neural network approach in this framework. The AR(3) Linear Model emerges as the most reliable and stable strategy, validating the importance of systematic backtesting and fair model comparison under identical trading conditions.





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